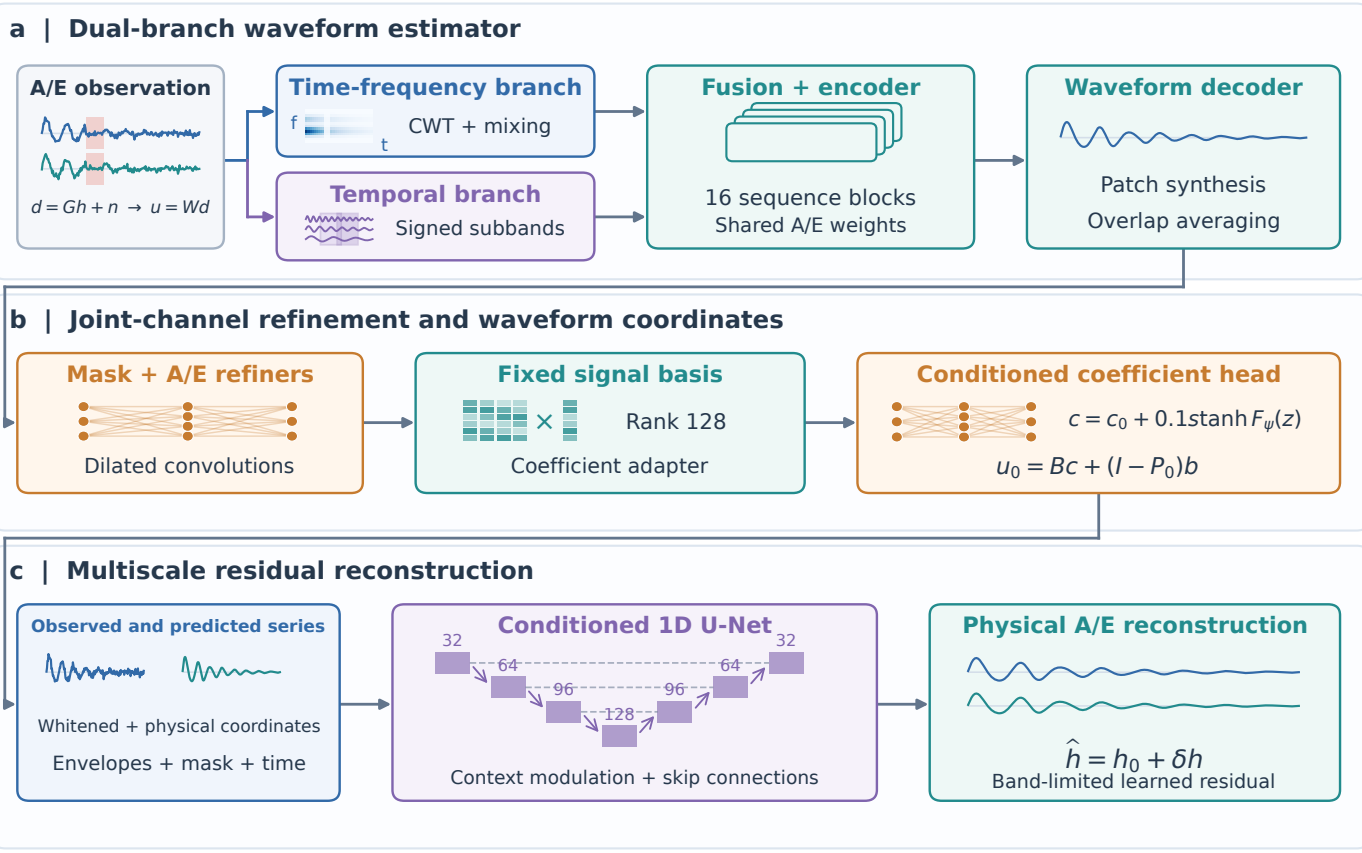

a | Dual-branch waveform estimator
A/E observation
$d = Gh + n \rightarrow u = Wd$
Time-frequency branch
f
t
CWT + mixing
Temporal branch
Signed subbands
Fusion + encoder
16 sequence blocks
Shared A/E weights
Waveform decoder
Patch synthesis
Overlap averaging
b | Joint-channel refinement and waveform coordinates
Mask + A/E refiners
Dilated convolutions
Fixed signal basis
Rank 128
Coefficient adapter
Conditioned coefficient head
$c = c_0 + 0.1 \tanh F_\psi(z)$
$u_0 = Bc + (I - P_0)b$
c | Multiscale residual reconstruction
Observed and predicted series
Whitened + physical coordinates
Envelopes + mask + time
Conditioned 1D U-Net
32
64
96
128
96
64
32
Context modulation + skip connections
Physical A/E reconstruction
$\hat{h} = h_0 + \delta h$
Band-limited learned residual